

Financial Frictions in Macroeconomics

Lecture Notes

QEM – Macroeconomics

Contents

1	Introduction: Financial Markets and Asymmetric Information	3
1.1	Agency Costs	3
2	The Romer (CSV) Model	4
2.1	Assumptions	4
2.2	Preliminary Implications	4
2.3	Key Assumption: Costly State Verification (CSV)	5
2.4	The Optimal Contract	5
2.4.1	Investor's Participation Constraint	5
2.4.2	Why the Optimal Contract is a Debt Contract	5
3	Equilibrium Value of D	6
3.1	Investor's Expected Net Receipts: $R(D)$	6
3.2	Properties of $R(D)$	7
3.3	Three Cases and Credit Rationing	7
3.4	Equilibrium D^*	7
4	Equilibrium Investment and Agency Costs	8
4.1	Agency Cost A	8
4.2	Entrepreneur's Investment Decision	8
5	The Financial Accelerator	9
5.1	Main Mechanism	9
5.2	Additional Channels	9
6	Empirical Evidence	10
7	A Macro Model with Financial Frictions	10
7.1	Overview	10

7.2	The Stochastic Discount Factor	11
7.3	Firm's Objective	11
7.4	Financing: Equity and Debt	12
7.5	Modigliani–Miller and the Need for Frictions	12
8	The Firm's Optimisation Problem	12
8.1	Three Frictions (Assumptions A1–A3)	12
8.2	Recursive Formulation	13
8.3	First-Order Conditions	14
8.4	Envelope Conditions	15
8.5	Optimality Conditions for Investment and Debt	15
8.6	Household Optimality Conditions	16
9	Two-Period Model	16
9.1	Functional Forms	16
9.2	Solution at $t = 2$	17
9.3	Equilibrium Equations at $t = 0$	17
10	Properties of the Model	18
10.1	Property 1: Positive Productivity Shock ($z \uparrow$ and $z' \uparrow$)	18
10.2	Property 2: Positive Asset Price Shock ($p \uparrow$) with $\mu > 0$ and $B > 0$	19
10.3	Property 3: Negative Financial Shock ($\xi \downarrow$) with $\mu > 0$	19
10.4	Property 4: Positive Financial Shock ($\xi \uparrow$) with $\mu = 0$ Initially	19
10.5	Property 5: Large Negative Financial Shock ($\xi \downarrow$) with $\mu = 0$ Initially	20
10.6	The Asymmetry of Financial Shocks	20
11	Key Conclusions	20

1 Introduction: Financial Markets and Asymmetric Information

In standard models (Chapters 8 and 9 of Romer), all agents are equally well-informed and financial markets function efficiently, achieving the Arrow–Debreu outcome. In such an ideal world, markets are complete: there exists a contract for every possible future state of the world, risk is perfectly allocated, and the First Welfare Theorem holds.

In reality, however, **firms are much better informed than outside investors** about their investment projects. Outside investors typically have little contact with the firm, little expertise regarding its activities, and insufficient stakes to justify acquiring relevant information. Even specialised intermediaries (banks, mutual funds, rating agencies) can be far less informed than the firms in which they invest.

This creates **asymmetric information**, which in turn gives rise to **agency problems** between investors and firms. Because investors bear some of the risk of the project — for instance, when the firm may go bankrupt — the firm can exploit its informational advantage. This generates **financial frictions**: wedges between the cost of internal and external finance that distort real economic outcomes.

1.1 Agency Costs

Agency costs are the costs that arise from the principal–agent problem between the firm and the outside investor. They manifest through three channels:

1. **Distortion of investment choices.** The firm may choose projects that are not socially optimal. For example, it may choose excessively risky projects, knowing that if the project fails, part of the loss is absorbed by the investor (limited liability). The investor cannot prevent this because they cannot perfectly observe the firm's decisions.
2. **Monitoring expenditure.** Since the investor cannot trust the firm blindly, they devote real resources to monitoring the firm's activities: audits, contractual covenants, due diligence. These resources represent a deadweight loss to society — they produce nothing and exist solely because of informational asymmetry.
3. **Reduced managerial effort (moral hazard).** Managers may exert less than the socially optimal level of effort. Because the outside investor captures part of the project's return, the firm does not fully internalise the benefits of its own effort. Since effort is unobservable to the investor, it cannot be directly contracted upon.

All three channels raise the cost of external finance above internal finance, generating an

external finance premium.

2 The Romer (CSV) Model

2.1 Assumptions

1. An investment project requires **1 unit of resources**.
2. The entrepreneur has wealth $W < 1$. Therefore, they must obtain $1 - W$ units of **outside financing**.
3. Expected output is $\gamma > 0$, which is **heterogeneous across entrepreneurs** and **publicly observable**. Actual output is distributed uniformly:

$$\text{output} \sim U[0, 2\gamma], \quad \Rightarrow \quad \mathbb{E}[\text{output}] = \gamma$$

4. Both the entrepreneur and outside investors are **risk-neutral**: they maximise expected payoffs without concern for variance.
5. There exists a **safe technology** with a certain rate of return r . This is the opportunity cost of capital for both parties.
6. Outside investors are **competitive**: in equilibrium, their expected return equals r .

Note on limited liability: Since the entrepreneur's entire wealth is invested in the project, their payment to outside investors cannot exceed the project's output. This forces investors to bear part of the project's risk.

2.2 Preliminary Implications

From these assumptions, two key conditions follow:

- **Social desirability:** The project is socially desirable if and only if

$$\gamma > 1 + r$$

The expected return must exceed the opportunity cost of the 1 unit of resources deployed.

- **Entrepreneur's participation condition:** The entrepreneur undertakes the project if and only if

$$\gamma - \mathbb{E}[\text{payment to investor}] > (1 + r)W$$

What the entrepreneur keeps must exceed what they would earn by investing their wealth W in the safe technology.

2.3 Key Assumption: Costly State Verification (CSV)

The central friction of the model (Townsend, 1979):

- The entrepreneur observes their output **costlessly**.
- An outside investor must pay a cost c to observe output, where $0 < c < \gamma$.

This asymmetry gives the entrepreneur an incentive to misreport output. Without verification, the investor cannot distinguish truth from lies. Verification is costly and therefore introduces an agency cost into the contract.

2.4 The Optimal Contract

2.4.1 Investor's Participation Constraint

Since the investor is competitive and risk-neutral, their expected receipts *net of verification costs* must equal their opportunity cost:

$$\mathbb{E}[\text{payment to investor}] - \mathbb{E}[\text{verification costs}] = (1 + r)(1 - W)$$

Since the investor must be compensated for expected verification expenditure, the entrepreneur's expected income becomes:

$$\mathbb{E}[\text{entrepreneur's income}] = \gamma - (1 + r)(1 - W) - A$$

where $A \equiv \mathbb{E}[\text{verification costs}]$ is the agency cost. To maximise their own payoff, the entrepreneur wants to **minimise** A , i.e., minimise the fraction of time the investor verifies output.

2.4.2 Why the Optimal Contract is a Debt Contract

The optimal contract that minimises verification is a **standard debt contract**. This is not assumed — it is derived from five steps:

1. **Without verification, the payment cannot depend on output.** If the contract specifies payment Q_1 for output Y_1 and $Q_2 > Q_1$ for output Y_2 , without verification the entrepreneur will always report Y_1 and pay Q_1 . Therefore, unverified payments must be constant.

2. **The payment with verification must be strictly below D .** If verified payment exceeded D , the entrepreneur would always pretend output is not in the verified region, avoiding the higher payment. Furthermore, it cannot equal D exactly: if the entrepreneur can pay D under verification, the investor can simply accept D without verifying, saving cost c .
3. **When output $\geq D$, the payment is exactly D .** If it were less than D for some output above D , the investor could raise the payment to D in those states, increasing expected receipts without raising verification costs — contradicting optimality.
4. **The entrepreneur cannot pay D when output $< D$.** By definition, output is insufficient to cover D , so bankruptcy (and hence verification) is unavoidable.
5. **In bankruptcy, the investor receives all of output.** If the payment were less than output in bankruptcy states, raising it would increase the investor's expected receipts without changing verification frequency, allowing a reduction in D and a saving on verification costs.

Together, these steps imply the unique optimal contract:

$$\text{Payment to investor} = \begin{cases} D & \text{if output} \geq D \\ \text{output} & \text{if output} < D \end{cases}$$

The entrepreneur borrows $1 - W$ and promises to repay D . If output is sufficient, they repay D and keep the surplus. If not, all resources go to the lender and the investor pays the verification cost c .

3 Equilibrium Value of D

3.1 Investor's Expected Net Receipts: $R(D)$

Assuming output $\sim U[0, 2\gamma]$, the investor's expected receipts minus verification costs are:

$$R(D) = \begin{cases} \frac{2\gamma - D}{2\gamma} \cdot D + \frac{D}{2\gamma} \left(\frac{D}{2} - c \right) & \text{if } D \leq 2\gamma \\ \gamma - c & \text{if } D > 2\gamma \end{cases} \quad (10.8)$$

Derivation of each term (case $D \leq 2\gamma$):

- **Good state** (output $\geq D$): probability = $\frac{2\gamma - D}{2\gamma}$; investor receives D ; no verification cost. Contribution: $\frac{2\gamma - D}{2\gamma} \cdot D$.

- **Bankruptcy** (output $< D$): probability = $\frac{D}{2\gamma}$; conditional expected output = $\frac{D}{2}$ (midpoint of $U[0, D]$); investor receives output but pays c . Contribution: $\frac{D}{2\gamma} \left(\frac{D}{2} - c\right)$.

The case $D > 2\gamma$ is a limiting case included for mathematical completeness: if D exceeds the maximum possible output, bankruptcy always occurs and $R(D) = \gamma - c$.

3.2 Properties of $R(D)$

Differentiating $R(D)$ for $D \leq 2\gamma$:

$$R'(D) = 1 - \frac{c}{2\gamma} - \frac{D}{2\gamma}$$

Setting $R'(D) = 0$:

$$D^* = 2\gamma - c \equiv D^{\text{MAX}}$$

$R(D)$ is **concave** in D (coefficient on D^2 is negative), increasing until $D = 2\gamma - c$ and decreasing thereafter. The maximum expected net revenue is:

$$R^{\text{MAX}} = R(2\gamma - c) = \left(\frac{2\gamma - c}{2\gamma}\right)^2 \gamma$$

Note: when $c = 0$, $R^{\text{MAX}} = \gamma$ (no friction, investor can capture full expected output). When $c > 0$, $R^{\text{MAX}} < \gamma$.

3.3 Three Cases and Credit Rationing

The equilibrium condition is $R(D^*) = (1 + r)(1 - W)$. Three scenarios arise:

1. **Unique equilibrium:** $(1 + r)(1 - W) < \gamma - c \Rightarrow$ one value of D solves $R(D) = (1 + r)(1 - W)$.
2. **Two solutions:** $(1 + r)(1 - W)$ between $\gamma - c$ and $R^{\text{MAX}} \Rightarrow$ two values $D_2^A < D_2^B$ solve $R(D) = (1 + r)(1 - W)$. Competition among investors drives D down to D_2^A (the lower solution), since any investor offering D_2^B can be undercut.
3. **Credit rationing:** $(1 + r)(1 - W) > R^{\text{MAX}} \Rightarrow$ no value of D satisfies the investor's participation constraint. Investors refuse to lend at any interest rate.

3.4 Equilibrium D^*

Solving $R(D) = (1 + r)(1 - W)$ and taking the smaller root:

$$\boxed{D^* = 2\gamma - c - \sqrt{(2\gamma - c)^2 - 4\gamma(1 + r)(1 - W)}} \quad (10.9)$$

for $(1+r)(1-W) \leq R^{\text{MAX}}$.

Note: $D^* > (1+r)(1-W)$ always. The entrepreneur must promise more than the investor's opportunity cost to compensate for states where output falls short and verification costs arise.

4 Equilibrium Investment and Agency Costs

4.1 Agency Cost A

The investor verifies output when output $< D^*$, which occurs with probability $\frac{D^*}{2\gamma}$. Expected verification costs are:

$$A = \frac{D^*}{2\gamma} \cdot c \quad (10.10)$$

Substituting the expression for D^* :

$$A = \left[\frac{2\gamma - c}{2\gamma} - \sqrt{\left(\frac{2\gamma - c}{2\gamma} \right)^2 - \frac{(1+r)(1-W)}{\gamma}} \right] c$$

Direct differentiation yields:

$$A = A(c, r, W, \gamma), \quad A_c > 0, \quad A_r > 0, \quad A_W < 0, \quad A_\gamma < 0 \quad (10.11)$$

Intuition for each partial derivative:

- $A_c > 0$: higher verification cost raises agency costs directly.
- $A_r > 0$: higher r raises $(1+r)(1-W)$, requiring higher D^* , increasing bankruptcy probability and hence A .
- $A_W < 0$: higher W reduces external financing needs $(1-W)$, lowering D^* and bankruptcy probability.
- $A_\gamma < 0$: higher expected output reduces the relative likelihood of bankruptcy.

4.2 Entrepreneur's Investment Decision

The entrepreneur undertakes the project if and only if:

$$\gamma - (1+r)(1-W) - A > (1+r)W$$

which simplifies to:

$$\gamma > 1 + r + A$$

Without frictions ($c = 0$, so $A = 0$), this reduces to $\gamma > 1 + r$ — the social optimum. With frictions ($A > 0$), some socially desirable projects ($\gamma > 1 + r$) are not undertaken because A raises the effective hurdle rate. This is the central inefficiency of the model.

5 The Financial Accelerator

5.1 Main Mechanism

The central implication of the CSV model is that financial market imperfections **magnify the effects of shocks to the economy** — the **financial accelerator** (Bernanke, Gertler, Gilchrist, 1996).

Since $A_W < 0$, a fall in W raises A , making the investment condition $\gamma > 1 + r + A$ harder to satisfy. Projects that were previously viable are no longer undertaken. The full transmission chain is:

$$W \downarrow \Rightarrow (1 - W) \uparrow \Rightarrow (1 + r)(1 - W) \uparrow \Rightarrow D^* \uparrow \Rightarrow \frac{D^*}{2\gamma} \uparrow \Rightarrow A \uparrow \Rightarrow \text{investment} \downarrow$$

More broadly, W corresponds to the firm's **net worth** in the real economy:

$$W \approx \text{current revenues} - \text{current expenses}$$

A negative shock to the economy reduces revenues, lowers net worth, raises agency costs, reduces investment, and further lowers output — creating a self-reinforcing loop:

$$\text{shock} \rightarrow W \downarrow \rightarrow A \uparrow \rightarrow \text{investment} \downarrow \rightarrow \text{output} \downarrow \rightarrow W \downarrow \rightarrow \dots$$

5.2 Additional Channels

- **Collateral values:** A fall in asset prices reduces the value of collateral firms can pledge, tightening borrowing constraints even beyond the W channel.
- **Interest rate shocks:** Since $A_r > 0$, a rise in r raises agency costs *in addition to* the standard cost-of-capital effect, making monetary policy particularly powerful in models with financial frictions.

- **Other variables:** Tax rates, idiosyncratic risk, and regulatory changes can also shift A .
- **The financial system itself:** The parameter c captures the efficiency of the financial system. A more developed financial system (lower c) reduces agency costs and supports higher investment, both in the short run (cyclical) and the long run (growth).

6 Empirical Evidence

The theory predicts that **internal resources affect investment** beyond what interest rates and profitability alone would imply. There is an unusual degree of empirical consensus on this point:

- **Fazzari, Hubbard, and Petersen (1988):** Firms that pay dividends have a margin of adjustment — they can reduce dividends to free up internal funds for investment. These firms should be less sensitive to financial frictions. Consistent with theory, they find investment is less sensitive to cash flow for dividend-paying firms.
- **Lamont (1997):** Studied conglomerates with both oil and non-oil divisions. When oil prices fell, the oil division generated less cash flow, reducing the entire conglomerate's net worth W . The non-oil divisions — whose projects were unrelated to oil — also invested less. This provides clean evidence for the $W \rightarrow A \rightarrow$ investment channel.
- **Collateral approach:** If financial frictions are present, the value of collateral assets should affect investment. Empirical work confirms this: firms with more collateral (e.g., real estate) invest more, especially when asset prices rise.

7 A Macro Model with Financial Frictions

7.1 Overview

We now embed financial frictions in a general equilibrium model (Quadrini, “Credit Market Frictions”; PhD Macro Book, Ch. 19). The goal is to formalise how frictions shape the transmission of both non-financial and financial shocks to the rest of the economy.

Starting point: neoclassical model

- **Representative household:** works, consumes, saves, and lends to firms.

- **Firms:** solve a dynamic optimisation problem. Investment today increases capital stock and profits in the future.
- Firms borrow from households; households lend their savings to firms.

7.2 The Stochastic Discount Factor

The firm is owned by the representative household. Therefore, the firm values future dividends using **the household's marginal utility**, not an arbitrary discount rate.

Step 1: A unit increase in consumption at $t > 0$ yields utility:

$$u_1(C_t, L_t)$$

where u_1 is the marginal utility of consumption. Since u is concave ($u_{11} < 0$), marginal utility is decreasing: the more the household consumes, the less it values one additional unit.

Step 2: The discounted value of that utility gain, brought back to period 0, is:

$$\beta^t u_1(C_t, L_t)$$

where $\beta < 1$ captures the household's **impatience** — it prefers the present to the future.

Step 3: Define Δ as the increase in consumption at $t = 0$ that generates the same utility as one unit at period t :

$$\Delta \cdot u_1(C_0, L_0) = \beta^t u_1(C_t, L_t)$$

Solving:

$$\Delta = \frac{\beta^t u_1(C_t, L_t)}{u_1(C_0, L_0)} \equiv m_t$$

m_t is the **stochastic discount factor (SDF)**: the number of units of consumption today that are equivalent (in utility terms) to one unit of consumption at period t . It is “stochastic” because future consumption C_t is uncertain, so m_t is a random variable.

7.3 Firm's Objective

The firm maximises the present value of all future dividends, weighted by the household's SDF:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} m_t d_t \right]$$

where d_t are dividends paid at time t .

7.4 Financing: Equity and Debt

The firm has two sources of financing:

- **Equity:** The household owns the firm and receives dividends d_t . Equity represents ownership — if profits are high, dividends are high; if profits are low, dividends are low. The return is variable.
- **Non-contingent debt:** The firm borrows $\frac{b_{t+1}}{R_t}$ from households at time t and promises to repay b_{t+1} at time $t + 1$, regardless of outcomes. R_t is the gross interest rate; the difference between what is borrowed and what is repaid is the interest cost.

7.5 Modigliani–Miller and the Need for Frictions

In the frictionless neoclassical model, **Modigliani–Miller (MM) holds:** the firm’s financing structure (debt vs. equity) is irrelevant to its value and real decisions. Debt and equity are perfect substitutes.

To make financial decisions matter for real outcomes (investment, output), frictions must be introduced.

Assumption A1: The cost of debt is lower than the cost of equity.

Possible reasons:

- **Tax advantage of debt:** Interest payments are tax-deductible, while dividends are not.
- **Convenience yield:** Debt (bonds) is highly liquid and can be used as collateral, giving it extra value beyond its return.

Formally, the effective cost of debt is:

$$\tilde{R}_t = \frac{R_t}{1 + \tau}$$

where $\tau > 0$ is the tax subsidy on debt. Since $\tilde{R}_t < R_t$, debt is cheaper than equity, breaking MM and making the financing mix relevant for real decisions.

8 The Firm’s Optimisation Problem

8.1 Three Frictions (Assumptions A1–A3)

The model introduces three frictions that break Modigliani–Miller and make financial decisions matter for real outcomes:

- **A1 — Debt is cheaper than equity.** The effective cost of debt is $\tilde{R}_t = R_t/(1+\tau)$, where $\tau > 0$ is the tax subsidy. The household lends at R_t ; the government subsidises the difference, so the firm's net borrowing cost is $\tilde{R}_t < R_t$.
- **A2 — Collateral constraint.** The firm's borrowing is limited by the value of its capital:

$$\frac{b_{t+1}}{\tilde{R}_t} \leq \xi p_t k_{t+1}$$

where $\xi \in (0, 1)$ is the fraction of capital that can be pledged as collateral and p_t is the price of capital. This constraint does not apply to equity financing. The price p_t is exogenous to the individual firm but depends on aggregate shocks — a key amplification channel.

- **A3 — Dividend smoothing cost.** Firms have a target dividend level ϕ and incur a quadratic cost when actual dividends deviate from it:

$$K(d_t - \phi)^2$$

This captures the empirical regularity that firms smooth dividends over time. It implies that the total resource cost of paying d_t to shareholders is:

$$\varphi(d_t) = d_t + K(d_t - \phi)^2$$

When A3 is costless ($K = 0$), the firm can freely adjust dividends to finance investment. With $K > 0$, this channel is restricted, making external financing more important.

8.2 Recursive Formulation

The firm's problem written in recursive (Bellman equation) form is:

$$\Omega(S; k, b) = \max_{d, \ell, k', b'} \{d + \mathbb{E} m' \Omega(S'; k', b')\} \quad (19.3)$$

subject to the **budget constraint**:

$$F(z, k, \ell) - w\ell + \frac{b'}{\tilde{R}} = b + p[k' - (1 - \delta)k] + \varphi(d)$$

and the **collateral constraint**:

$$\xi p k' \geq \frac{b'}{\tilde{R}}$$

Notation and interpretation:

- $\Omega(S; k, b)$: value of the firm — the expected discounted sum of all future dividends.
- *State variables*: S (aggregate state of the economy), k (capital inherited from last period), b (debt inherited from last period). These cannot be changed today.
- *Control variables*: d (dividends today), ℓ (labour hired), k' (capital next period), b' (new debt).
- *Budget constraint (sources = uses)*: left-hand side = operating profits $F(z, k, \ell) - w\ell$ plus new debt proceeds $b'/\tilde{R}$; right-hand side = repayment of old debt b , net investment $p[k' - (1 - \delta)k]$, and dividend payment $\varphi(d)$.
- $k' - (1 - \delta)k$: net new investment. The firm starts with k units of capital, of which $(1 - \delta)k$ survive depreciation. To reach k' next period, it must purchase $k' - (1 - \delta)k$ new units at price p .
- The discount factor m' is the household's SDF. The firm uses it because it is owned by the household. Since the firm is atomistic, it takes m' as given.

8.3 First-Order Conditions

The Lagrangian is:

$$\mathcal{L} = d + \mathbb{E} m' \Omega(S'; k', b') + \lambda \left[F(z, k, \ell) - w\ell + \frac{b'}{\tilde{R}} - b - p(k' - (1 - \delta)k) - \varphi(d) \right] + \mu \lambda \left(\xi p k' - \frac{b'}{\hat{R}} \right)$$

where λ is the multiplier on the budget constraint and $\mu\lambda$ is the multiplier on the collateral constraint. When $\mu > 0$, the collateral constraint is **binding** — the firm would like to borrow more but cannot.

The FOCs are:

$$\frac{\partial \mathcal{L}}{\partial d} = 1 - \lambda \varphi'(d) = 0 \quad \Rightarrow \quad \lambda = \frac{1}{\varphi'(d)} \quad (\text{FOC-}d)$$

$$\frac{\partial \mathcal{L}}{\partial \ell} = F_3(z, k, \ell) - w = 0 \quad (\text{FOC-}\ell)$$

$$\frac{\partial \mathcal{L}}{\partial k'} = \mathbb{E} m' \Omega_2(S'; k', b') - \lambda p + \mu \lambda \xi p = 0 \quad (\text{FOC-}k')$$

$$\frac{\partial \mathcal{L}}{\partial b'} = \mathbb{E} m' \Omega_3(S'; k', b') + \frac{\lambda}{\tilde{R}} - \frac{\mu \lambda}{\hat{R}} = 0 \quad (\text{FOC-}b')$$

Interpretation of FOC- d : $\lambda = 1/\varphi'(d)$ is the shadow value of one additional unit of resources for the firm. It increases when $\varphi'(d)$ is small, i.e., when dividends are far from

the target ϕ .

Interpretation of FOC- ℓ : The marginal product of labour equals the wage — identical to the frictionless model. Financial frictions do *not* directly distort labour demand.

8.4 Envelope Conditions

The envelope conditions give the marginal value of the state variables:

$$\Omega_2(S; k, b) = \lambda [F_2(z, k, \ell) + p(1 - \delta)]$$

$$\Omega_3(S; k, b) = -\lambda$$

Updated one period forward:

$$\Omega_2(S'; k', b') = \lambda' [F_2(z', k', \ell') + p'(1 - \delta)], \quad \Omega_3(S'; k', b') = -\lambda'$$

where $\lambda' = 1/\varphi'(d')$.

Intuition: $\Omega_2 > 0$ — one more unit of capital inherited raises firm value because it produces output (F_2) and retains market value ($p(1 - \delta)$). $\Omega_3 < 0$ — one more unit of inherited debt reduces firm value because it must be repaid.

8.5 Optimality Conditions for Investment and Debt

Substituting the envelope conditions into FOC- k' and FOC- b' , and using $\lambda'/\lambda = \varphi'(d)/\varphi'(d')$:

Investment (equation 19.5):

$$\mathbb{E} m' \frac{\varphi'(d)}{\varphi'(d')} [(1 - \delta)p' + F_2(z', k', \ell')] = (1 - \mu\xi)p \quad (19.5)$$

Debt (equation 19.6):

$$\mathbb{E} m' \frac{\varphi'(d)}{\varphi'(d')} \hat{R} = 1 - \mu \quad (19.6)$$

Interpretation of (19.5):

- *Left-hand side:* expected discounted gross return of one unit of capital — it produces F_2 and retains $(1 - \delta)p'$ in value.
- *Right-hand side:* effective cost of capital, $(1 - \mu\xi)p$. When the collateral constraint is binding ($\mu > 0$), the effective cost is **lower** than the market price p . This is the

collateral premium: each unit of capital allows the firm to borrow ξp more at the cheaper debt rate $\tilde{R}$, reducing the net cost of acquiring that unit.

Interpretation of (19.6): When the collateral constraint binds ($\mu > 0$), the firm would like to borrow more but cannot. The right-hand side $(1 - \mu) < 1$ captures the shadow cost of the constraint.

8.6 Household Optimality Conditions

For completeness, the representative household's optimality conditions are:

Labour supply:

$$u_1(C, L) w = -u_2(C, L)$$

The marginal utility gain from working (earning w and consuming it) equals the marginal disutility of labour.

Optimal bond holdings (Euler equation):

$$\frac{1}{R} = \mathbb{E} \left[\frac{\beta u_1(C', L')}{u_1(C, L)} \right]$$

The bond price equals the expected stochastic discount factor. This is the standard consumption Euler equation.

Optimal equity holdings:

$$q = \mathbb{E} \left[\frac{\beta u_1(C', L')}{u_1(C, L)} (d' + q') \right]$$

The equity price today equals the discounted expected value of tomorrow's dividend plus the equity price — the standard asset pricing equation.

9 Two-Period Model

9.1 Functional Forms

To make the model tractable, we adopt:

$$u(c_t, l_t) = \ln(c_t - l_t^\nu), \quad F(z_t, k_t, l_t) = z_t k_t^\alpha l_t^{1-\alpha}$$

where z_t is Total Factor Productivity (TFP), and α is the capital share. There is no uncertainty. The time horizon is $t = 0, 1, 2$, with $k_2 = 0$ and $b_2 = 0$ (no capital or debt carried beyond the final period).

Since the model has a finite horizon, we solve **backwards** from $t = 2$.

9.2 Solution at $t = 2$

Labour: The labour market condition equates the marginal product of labour to the marginal rate of substitution:

$$(1 - \alpha)z' \left(\frac{K'}{L'} \right)^\alpha = \nu L'^{\nu-1}$$

Solving:

$$L' = \left(\frac{1 - \alpha}{\nu} \right)^{\frac{1}{\alpha+\nu-1}} z'^{\frac{1}{\alpha+\nu-1}} K'^{\frac{\alpha}{\alpha+\nu-1}}$$

Both higher TFP ($z' \uparrow$) and more capital ($K' \uparrow$) raise labour productivity, increasing labour demand: $L' \uparrow$.

Output: $Y' = z' K'^\alpha L'^{1-\alpha}$ increases with both z' and K' .

Consumption: Since $t = 2$ is the final period, all available resources are consumed:

$$C' = (1 - \delta)K' + Y'$$

Utility: Define $g(z', K') \equiv C' - L'^\nu$. Then:

$$u(C', L') = \ln(g(z', K'))$$

Flow utility is increasing in both z' and K' : higher productivity and more capital raise output, consumption, and welfare.

9.3 Equilibrium Equations at $t = 0$

Assuming the collateral constraint is binding ($\mu > 0$), the model at $t = 0$ is characterised by six equations in six unknowns (K', B', C, D, R, μ):

$$(1 - \mu\xi)p = \beta \frac{C - L^\nu}{g(z', K')} [1 - \delta + F_2(z', K', L')] [1 + 2K(D - \phi)] \quad (1)$$

$$\frac{(1 + \tau)B'}{R} = \xi p K' \quad (2)$$

$$\alpha F(z, K, L) + \frac{(1 + \tau)B'}{R} = B + p(K' - (1 - \delta)K) + D + K(D - \phi)^2 \quad (3)$$

$$D + B + (1 - \alpha)F(z, K, L) = C + \frac{B'}{R} + T \quad (4)$$

$$\frac{(1 - \mu)(1 + \tau)}{R} = \beta \frac{C - L^\nu}{g(z', K')} [1 + 2K(D - \phi)] \quad (5)$$

$$\frac{1}{R} = \beta \frac{C - L^\nu}{g(z', K')} \quad (6)$$

Notes: Equations (5) and (6) together imply $(1 - \mu)(1 + \tau) = 1$, directly linking μ to the tax subsidy τ . Equations (3) and (4) are the budget constraints of the firm and household respectively.

10 Properties of the Model

10.1 Property 1: Positive Productivity Shock ($z \uparrow$ and $z' \uparrow$)

A persistent increase in TFP raises output in both periods: $Y \uparrow$ and $Y' \uparrow$.

From equation (1), $z' \uparrow \Rightarrow F_2(z', K', L') \uparrow \Rightarrow K' \uparrow$. Investment can be funded through:

- Higher current profits: $\alpha F(z, K, L) \uparrow$
- More borrowing capacity: $K' \uparrow \Rightarrow B' \uparrow$ (via the collateral constraint)

Dampening effect when constraint binds: If $\xi < 1$, not all investment can be funded with cheap debt. The firm must reduce dividends ($D \downarrow$) to finance additional investment with equity — but this is costly (A3). Hence, the investment response is **smaller** than in the frictionless benchmark. Having to fund investment with expensive equity raises the effective cost of capital and dampens the response to productivity shocks.

10.2 Property 2: Positive Asset Price Shock ($p \uparrow$) with $\mu > 0$ and $B > 0$

When the collateral constraint binds and the firm has outstanding debt, a rise in the capital price amplifies investment:

$$p \uparrow \Rightarrow \text{net worth} = pK - B \uparrow \Rightarrow K' \uparrow \Rightarrow Y' \uparrow$$

From equations (3) and (2), the equilibrium capital stock satisfies:

$$K' = \frac{1}{1 - \xi} \left((1 + \delta)K + \frac{\alpha F(z, K, L) - B - D - K(D - \phi)^2}{p} \right)$$

Higher p raises net worth directly, relaxing the collateral constraint and allowing more borrowing and investment. A larger ξ (firms can pledge more of their capital as collateral) amplifies this effect further — each unit of capital supports more debt, making the firm more sensitive to changes in p .

This is an amplification mechanism for real shocks: a rise in asset prices feeds into investment through the collateral channel, beyond the direct effect on the cost of capital.

10.3 Property 3: Negative Financial Shock ($\xi \downarrow$) with $\mu > 0$

A tightening of the collateral constraint reduces investment both directly and indirectly:

$$\xi \downarrow \xrightarrow{(2)} B' \downarrow \xrightarrow{(3)} D \downarrow \text{ and/or } K' \downarrow$$

Suppose $D \downarrow$. From equation (5), $\mu \uparrow$ — the shadow value of relaxing the constraint rises. From equation (1), the collateral benefit $\mu\xi$ may partially offset the fall in ξ , but the net effect is that $F_2(z', K', L')$ must rise to restore equilibrium, which requires $K' \downarrow$ (since the marginal product of capital is decreasing). Therefore:

$$\xi \downarrow \Rightarrow K' \downarrow \Rightarrow Y' \downarrow \Rightarrow L' \downarrow$$

10.4 Property 4: Positive Financial Shock ($\xi \uparrow$) with $\mu = 0$ Initially

If the collateral constraint is not binding to begin with, relaxing it further has **no effect** on investment, output, or employment:

$$\xi \uparrow \quad (\mu = 0) \Rightarrow \text{no effect on } K', Y, Y', L, L'$$

The firm was already unconstrained — it did not want to borrow more at the margin.

10.5 Property 5: Large Negative Financial Shock ($\xi \downarrow$) with $\mu = 0$ Initially

Even if the constraint starts non-binding, a sufficiently large fall in ξ can make it binding:

$$\xi \downarrow \text{ (large enough)} \Rightarrow \mu > 0 \Rightarrow K' \downarrow \Rightarrow Y' \downarrow \Rightarrow L' \downarrow$$

10.6 The Asymmetry of Financial Shocks

Properties 4 and 5 together reveal a crucial asymmetry:

Shock	Constraint initially slack ($\mu = 0$)	Constraint initially binding ($\mu > 0$)
$\xi \uparrow$	No effect	Positive effect
$\xi \downarrow$ (small)	No effect	Negative effect
$\xi \downarrow$ (large)	Negative effect	Negative effect

Financial booms (positive ξ shocks when $\mu = 0$) leave the real economy unaffected. Financial busts (negative ξ shocks large enough to make $\mu > 0$) cause sharp contractions in investment and output. This **asymmetric feature of financial shocks** is a key reason why models with financial frictions have been used to understand the dynamics of financial booms and busts, including the 2008 Global Financial Crisis.

11 Key Conclusions

1. **Investment depends on more than profitability.** With financial frictions, investment also depends on the firm's debt level B , asset prices p , and collateral quality ξ . The firm's balance sheet matters for real outcomes.
2. **Capital has a double value.** When the collateral constraint binds, capital not only produces output but also enables cheaper borrowing. This *collateral premium* lowers the effective cost of capital below its market price.
3. **Shocks are amplified.** A negative shock to productivity or collateral quality reduces borrowing capacity, forces costly equity financing, and cuts investment beyond the direct effect of the shock. The financial system amplifies and propagates shocks.
4. **Financial shocks are asymmetric.** Positive financial shocks ($\xi \uparrow$) have no effect when the constraint is slack. Negative financial shocks ($\xi \downarrow$) can suddenly make the constraint binding, causing sharp contractions. Booms are muted; busts are severe.

5. **Financing structure matters — Modigliani–Miller fails.** Debt is cheaper than equity (A1), but borrowing is capped by collateral (A2) and dividend adjustment is costly (A3). When firms are forced into expensive equity financing, the effective cost of capital rises and investment falls.